# Cisco PE Node Recovery from Hardware/Environmental Faults (ASIC Memory Parity Errors, Line Card Crashes, Power Supply Redundancy Loss)

**Node:** MALL-PE-08 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_3266

## Description

The Failure Domain of Hardware/Environmental Faults, as observed in the Malleswaram Node (MALL-PE-08) with a Cisco PE role, can be attributed to a multitude of underlying factors. Specifically, ASIC memory parity errors can arise from intermittent faults within the Application-Specific Integrated Circuit (ASIC) responsible for managing packet forwarding and processing. These errors can lead to data corruption and, subsequently, line card crashes, which in turn can cause service disruptions. Furthermore, the loss of power supply redundancy poses a significant risk to network availability, as the failure of a single power supply unit can lead to a complete system failure. The Cisco PE equipment, operating at the edge of the network, is particularly susceptible to these failures due to its critical role in managing traffic aggregation and distribution. The complex interplay between hardware components, environmental factors, and software configurations necessitates a comprehensive approach to troubleshooting and resolution, ensuring that the integrity of the network is maintained.

## Pre-requisite Checks

- Dispatch remote hands to Malleswaram Node for on-site support
- Verify redundant link status to ensure no single points of failure
- Check environmental conditions (temperature, humidity) at the site

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC memory parity errors | High |
| Line card hardware failures | Medium |
| Power supply unit failure | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show platform, Expected Output: Platform details, including hardware and software versions, Description: This command provides an overview of the platform's hardware and software components, allowing for the identification of potential hardware faults or version mismatches that could contribute to the failure.
```

```
Command: show ip interface brief, Expected Output: List of interfaces with their status and IP addresses, Description: This command isolates issues in the data plane by displaying the status of all interfaces, helping to identify any that may be down or experiencing errors, which could be indicative of line card crashes or power supply issues.

Command: show environment, Expected Output: Environmental sensor readings (temperature, voltage, etc.), Description: This command provides insights into the environmental conditions that could be affecting the hardware, such as overheating or power supply issues, which are critical for diagnosing hardware/environmental faults.
```

## Resolution Steps

Step 1: Step: Replace Faulty Line Card, Command: hw-module slot shutdown; hw-module slot online, Description: This step involves shutting down the faulty line card to prevent further disruptions and then replacing it with a functional one, ensuring that the replacement card is properly seated and powered on. The 'hw-module' commands are used to manage the hardware modules in Cisco devices.

Step 2: Step: Reset Power Supply Redundancy, Command: power supply redundant mode, Description: If the power supply redundancy loss is due to a configuration issue, this command resets the power supply to its redundant mode, ensuring that if one power supply fails, the other can take over without interrupting system operation.

Step 3: Step: Update ASIC Firmware, Command: request platform software package install, Description: This step involves updating the firmware of the ASIC to the latest version available, which may resolve known issues causing memory parity errors. The 'request platform software package install' command is used to install software packages on Cisco devices.

## Post-Check Verification

```
Command: show platform, Expected Output: Updated platform details reflecting changes made, Description: After performing the resolution steps, this command verifies that the hardware issues have been resolved and that the platform is operating as expected.

Command: show ip interface brief, Expected Output: All interfaces should be up and operational, Description: This command confirms that all interfaces are now operational, indicating successful resolution of line card or power supply issues affecting the data plane.

Command: show environment, Expected Output: Environmental readings within acceptable ranges, Description: Verifies that environmental conditions are within acceptable limits, ensuring that the hardware is operating under safe conditions.
```

## Roll-back Procedures

- Step: Revert to Previous Line Card Configuration, Command: hw-module slot shutdown; hw-module slot online, Description: If the replacement line card does not resolve the issue or introduces new problems, this step involves removing the new line card and reinstalling the original one to return to the previous state.

- Step: Reconfigure Power Supply to Original Settings, Command: no power supply redundant mode, Description: If resetting power supply redundancy caused unforeseen issues, this command reverts the power supply configuration to its original state, ensuring that any changes made during troubleshooting are reversed.

- Step: Revert to Previous ASIC Firmware Version, Command: request platform software package install previous-version, Description: If the updated firmware introduces compatibility issues or does not resolve the original problem, this step involves reverting to the previous version of the ASIC firmware to

restore the system to its previous operational state.